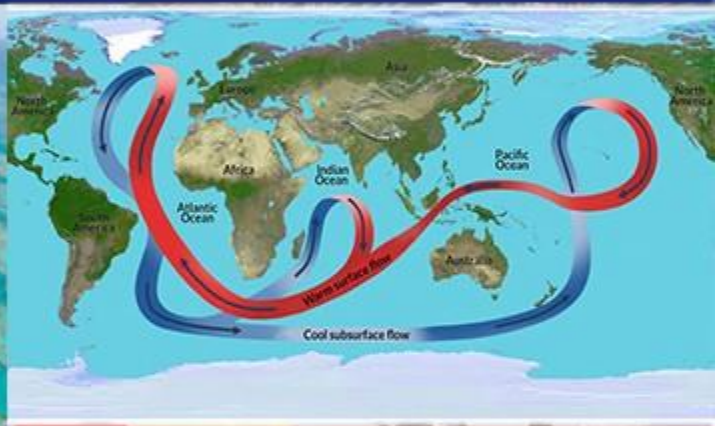


GEOGRAPHY

PHYSICAL GEOGRAPHY



OCEANOGRAPHY

MODULE - 4

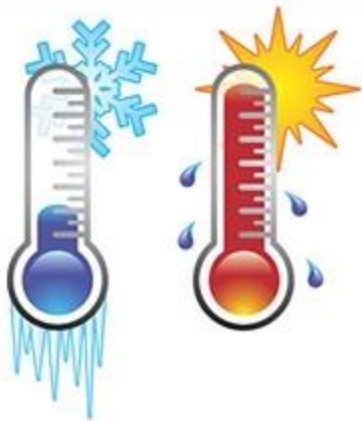
MOVEMENT OF
OCEAN WATER

MOVEMENTS OF OCEAN WATER

1 The ocean water is dynamic. Its physical characteristics like temperature, salinity, density and the external forces like that of the sun, moon and the winds influence the movement of ocean water.

1

TEMPERATURE



2

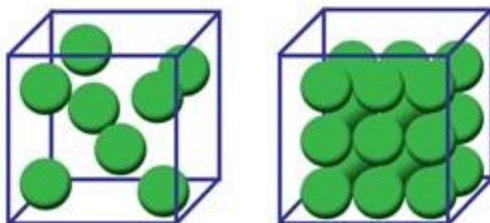
SALINITY



3

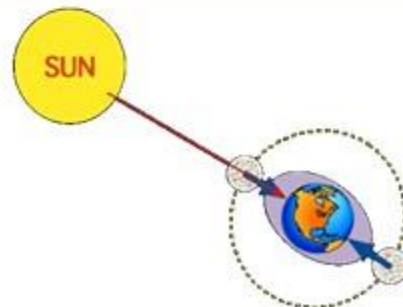
DENSITY

Density



4

FORCE OF SUN & MOON



5

WINDS

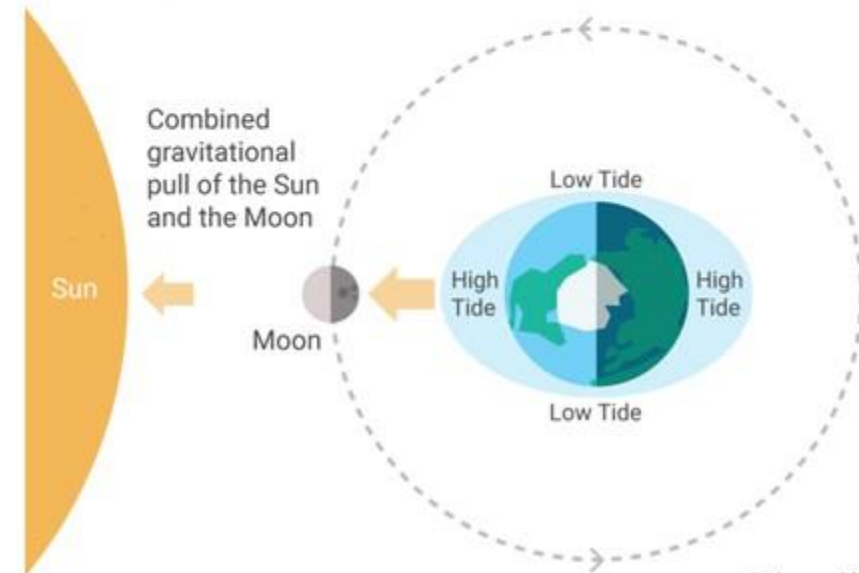
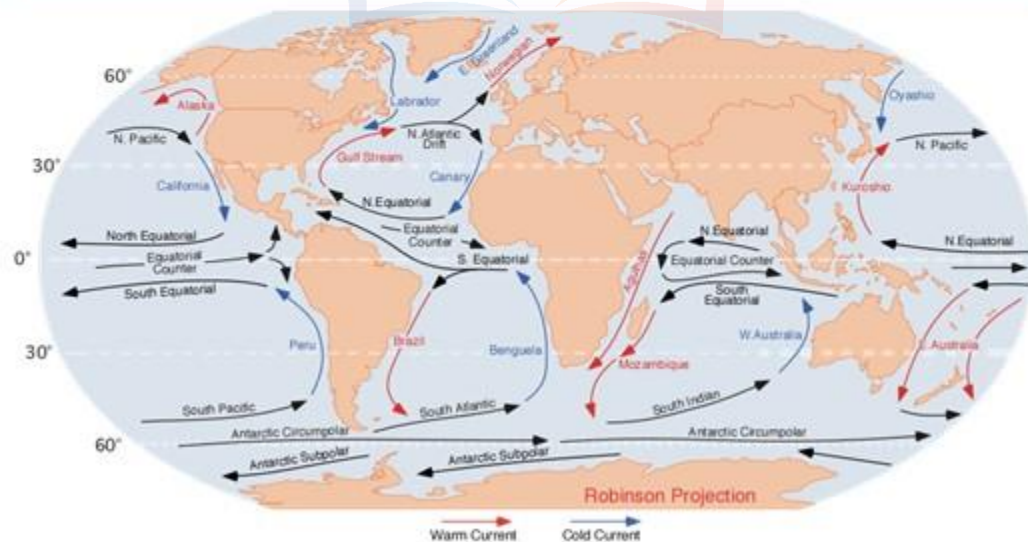


MOVEMENTS OF OCEAN WATER

2 The horizontal and vertical motions are common in ocean water bodies.

The horizontal motion refers to the **ocean currents and waves**.

The vertical motion refers to **tides**.



MOVEMENTS OF OCEAN WATER

MOVEMENTS OF OCEAN WATER

HORIZONTAL MOTION

- 1 Ocean currents are the **continuous flow of huge amount of water in a definite direction** while the waves are the release of energy in a horizontal motion.
- 2 Water moves ahead from one place to another through ocean currents while the water in the waves does not move, but the wave trains move ahead.

VERTICAL MOTION

- 1 The vertical motion refers to the rise and fall of water in the oceans and seas.
- 2 Due to attraction of the sun and the moon, the ocean water is raised up and falls down twice a day.
- 3 The upwelling of cold water from subsurface and the sinking of surface water are also forms of vertical motion of ocean water.

MOVEMENTS OF OCEAN WATER

WHAT DO YOU MEAN BY WAVES?



- 1 Waves are actually the energy, not the water as such, which moves across the ocean surface.
- 2 Water particles only travel in a small circle as a wave passes. Wind provides energy to the waves.
- 3 Wind causes waves to travel in the ocean and the energy gets released on shorelines.
- 4 The motion of the surface water seldom affects the stagnant deep bottom water of the oceans.

WAVES

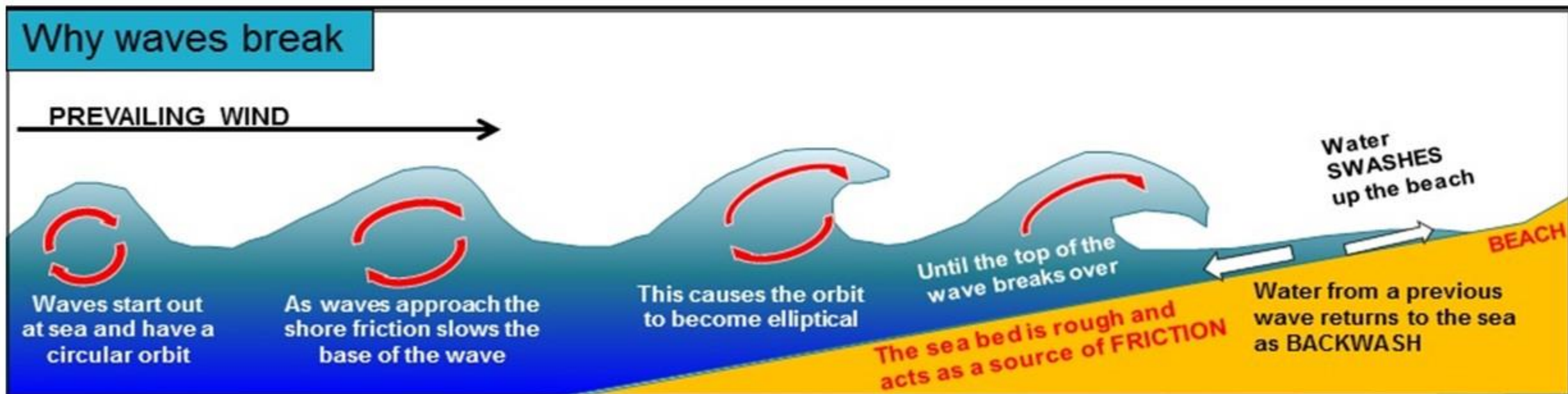
MOVEMENTS OF OCEAN WATER

WHAT WILL HAPPEN WHEN WAVES APPROACH THE COAST?

1 As a wave approaches the beach, it **slows down**. This is due to the **friction occurring** between the dynamic water and the sea floor.

2 And, when the depth of water is less than half the wavelength of the wave, the **wave breaks**.

Why waves break



WAVES

MOVEMENTS OF OCEAN WATER

WHAT WILL HAPPEN WHEN WAVES APPROACH THE COAST?

3 The largest waves are found in the open oceans.



4 Waves continue to grow larger as they move and absorb energy from the wind.

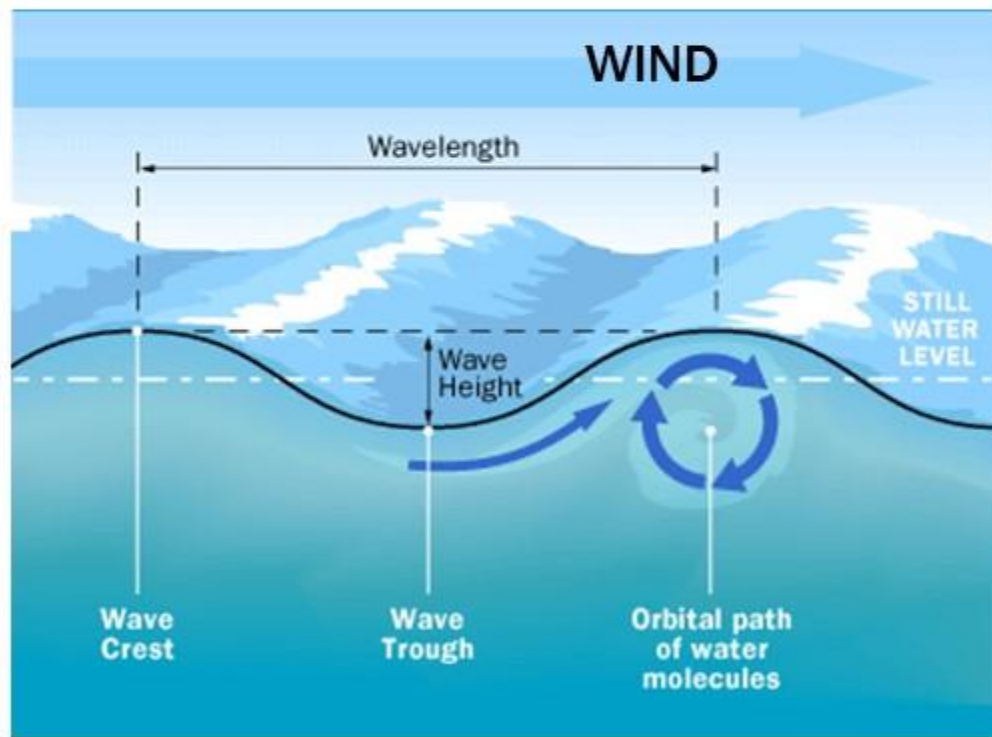


WAVES

MOVEMENTS OF OCEAN WATER

WHAT EFFECT WILL WINDS HAVE OVER WAVES?

- 1 Most of the waves are caused by the wind driving against water.



- 2 When a breeze blows over calm water, small ripples form and grow as the wind speed increases until white caps appear in the breaking waves.

- 3 Waves may travel thousands of kilometre before rolling ashore, breaking and dissolving as surf.

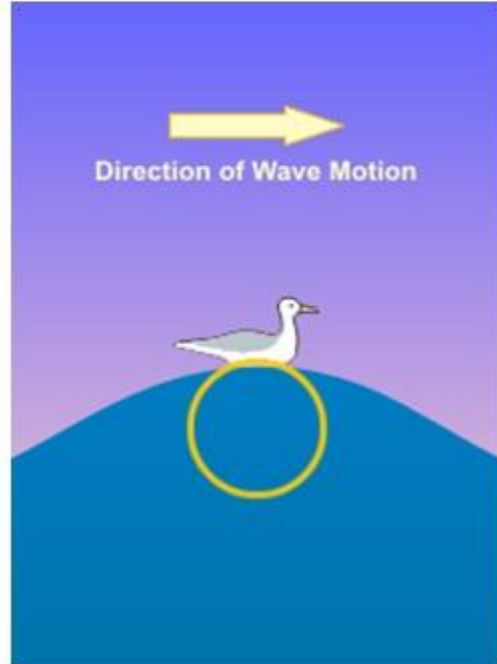
- 4 A wave's size and shape reveal its origin.
 - Steep waves are fairly young ones and are probably formed by local wind.
 - Slow and steady waves originate from far away places, possibly from another hemisphere.

WAVES

MOVEMENTS OF OCEAN WATER

WHAT EFFECT WILL WINDS HAVE OVER WAVES?

- 5 The maximum wave height is determined by the strength of the wind, i.e.
- how long it blows and
 - the area over which it blows in a single direction.



- 6 Waves travel because wind pushes the water body in its forward course while gravity pulls the crests of the waves downward.
- 7 The actual motion of the water beneath the waves is circular.
- 8 It indicates that things are carried up and forward as the wave approaches, and down and back as it passes.

MOVEMENTS OF OCEAN WATER

WAVES

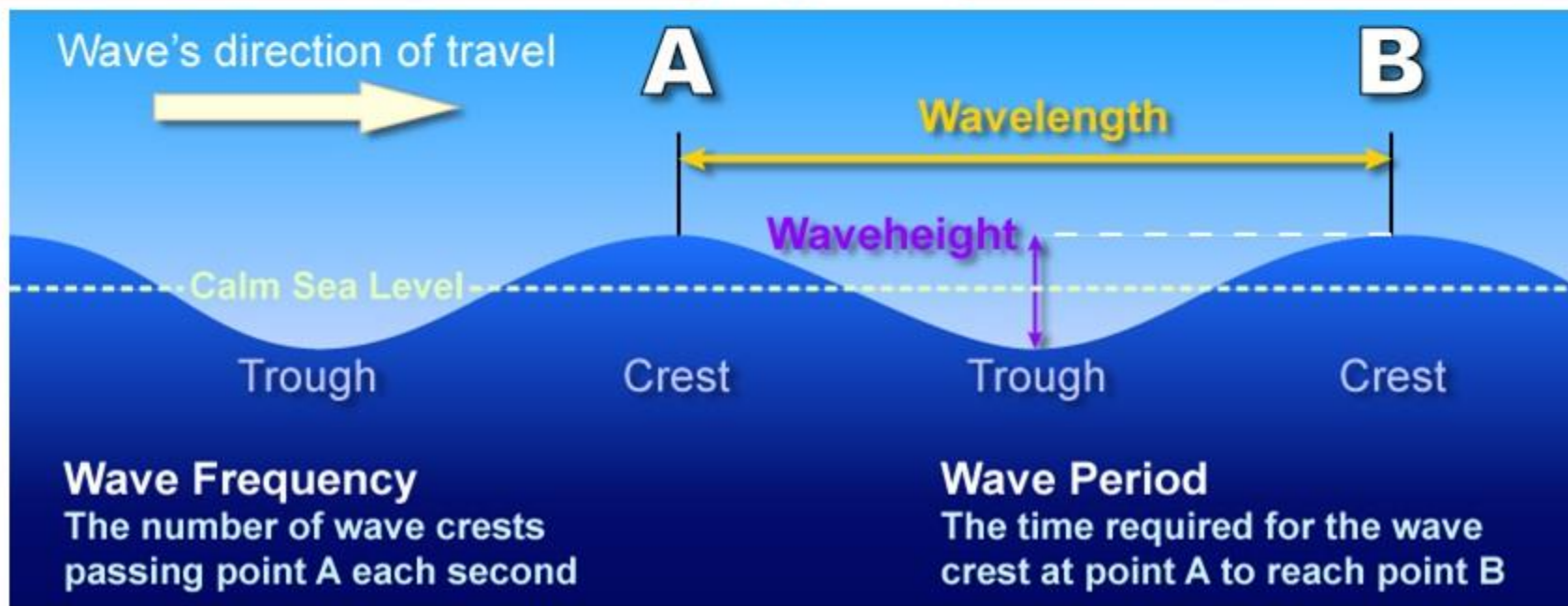
CHARACTERISTICS OF WAVES

1 Wave crest and trough : The **highest and lowest points of a wave** are called the crest and trough respectively.

2 Wave height : It is the **vertical distance** from the bottom of a trough to the top of a crest of a wave.

3 Wavelength : It is the **horizontal distance** between two successive crests.

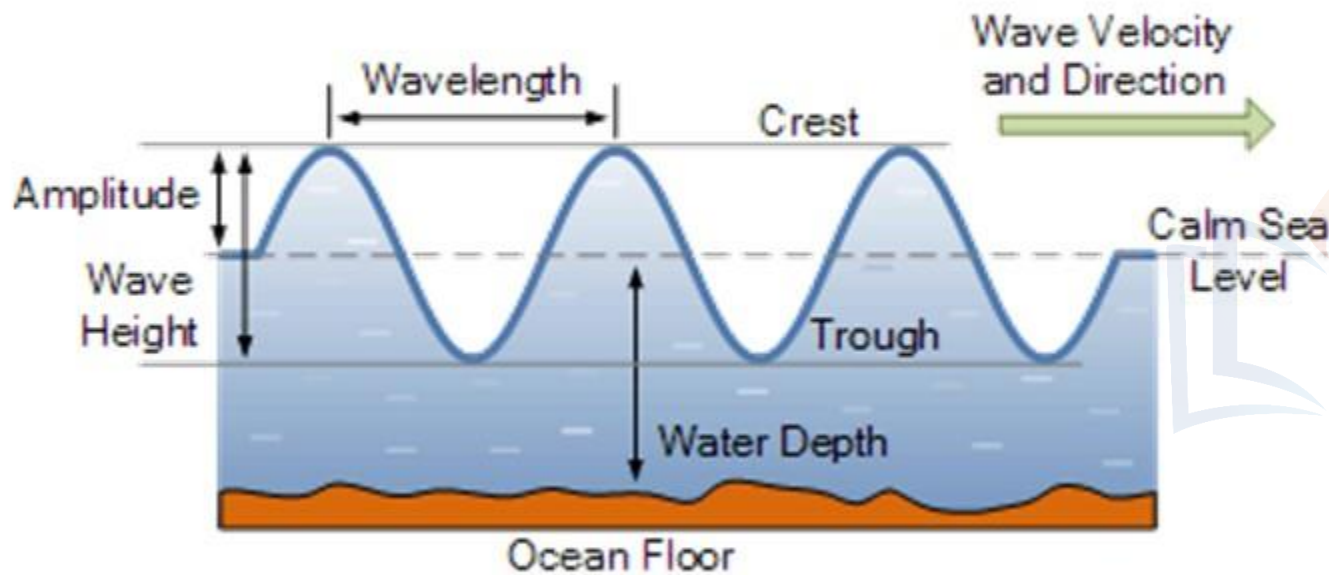
4 Wave period : It is the **time interval** between two successive wave crests or troughs as they pass a fixed point.



MOVEMENTS OF OCEAN WATER

WAVES

CHARACTERISTICS OF WAVES



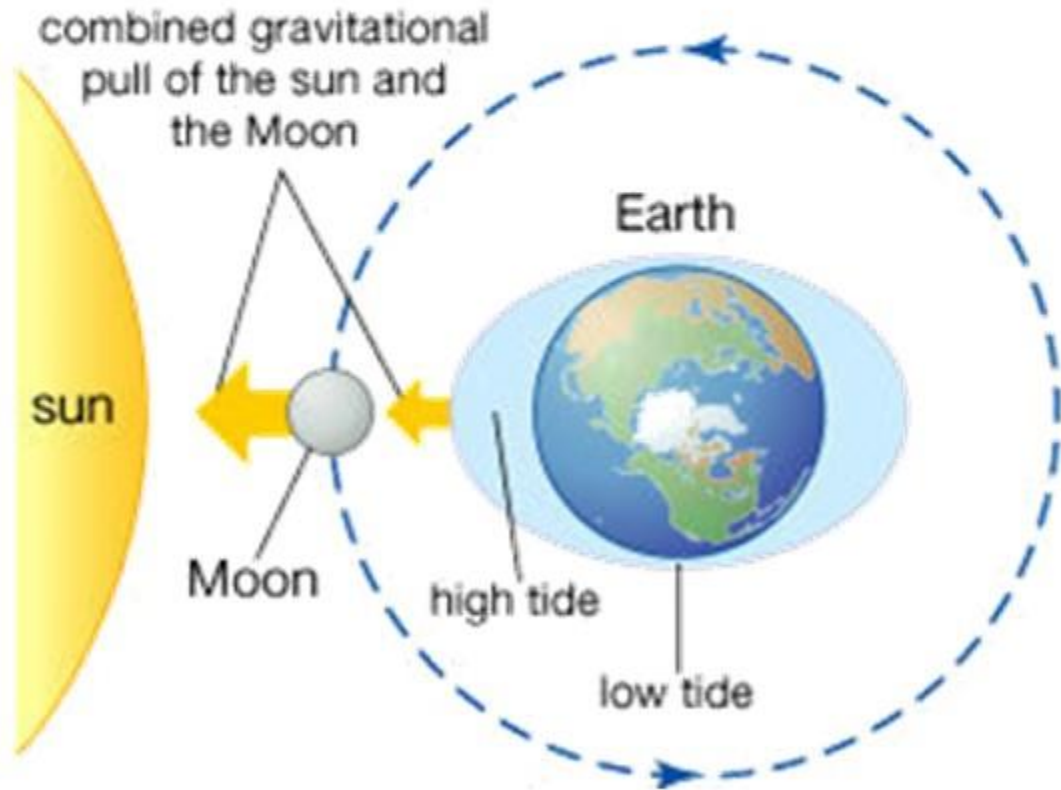
5 Wave amplitude : It is **one-half** of the wave height.

6 Wave speed : It is the **rate** at which the wave moves through the water, and is measured in knots.

7 Wave frequency : It is the **number** of **waves** passing a given point during a one second time interval.

MOVEMENTS OF OCEAN WATER

WHAT DO YOU MEAN BY TIDES?



1 The periodical rise and fall of the sea level, once or twice a day, mainly due to the attraction of the sun and the moon, is called a *tide*.

2 The study of tides is very complex, **spatially and temporally**, as it has great variations in frequency, magnitude and height.

SURGES ARE NOT TIDES!!

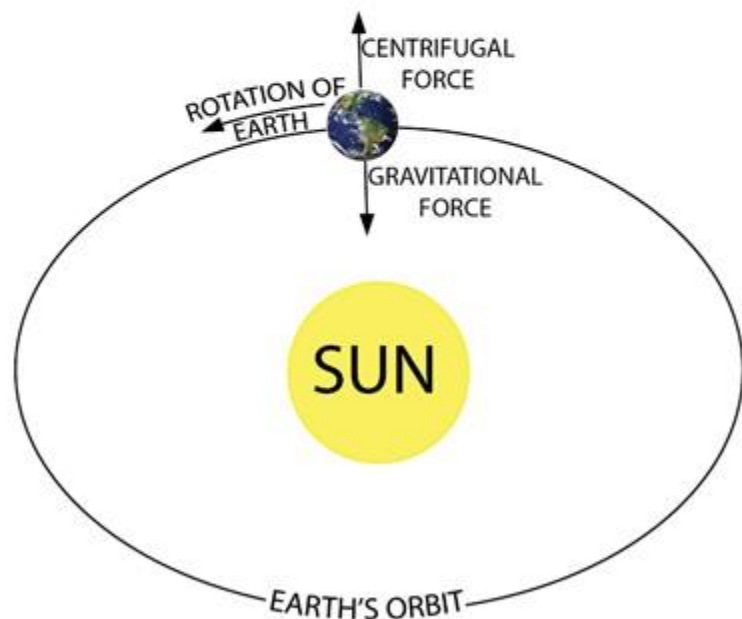
Movement of water caused by meteorological effects (winds and atmospheric pressure changes) are called *surges*. Surges are not regular like tides.

MOVEMENTS OF OCEAN WATER

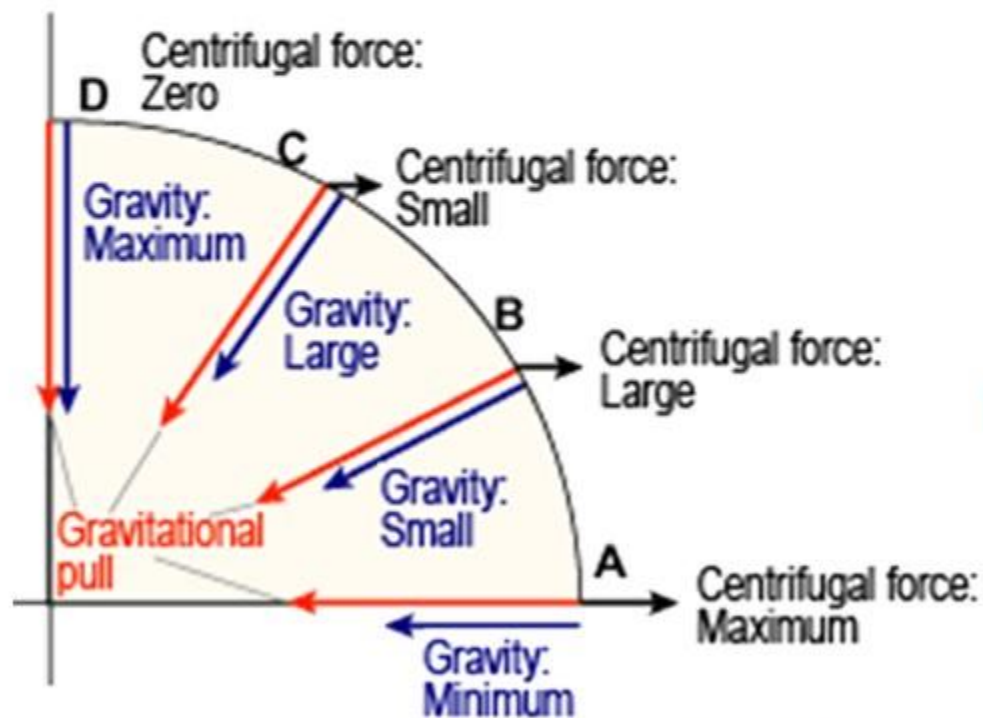
TIDES

FACTORS RESPONSIBLE FOR TIDES

- 1 The **moon's** gravitational pull to a great extent and to a lesser extent the **sun's** gravitational pull, are the major causes for the occurrence of tides.



- 2 Another factor is **centrifugal force**, which is the force that acts to **counter balance the gravity**.



Together, the gravitational pull and the centrifugal force are responsible for creating the two major tidal bulges on the earth.

MOVEMENTS OF OCEAN WATER

TIDES

MECHANISM OF TIDE FORMATION

1 On the side of the earth facing the moon, a tidal bulge occurs because of the gravitational attraction of the moon.

Why is it so?

2 On the surface of the earth, nearest to the moon, **pull or the attractive force of the moon is greater than the centrifugal force**, and so there is a net force causing a bulge towards the moon.

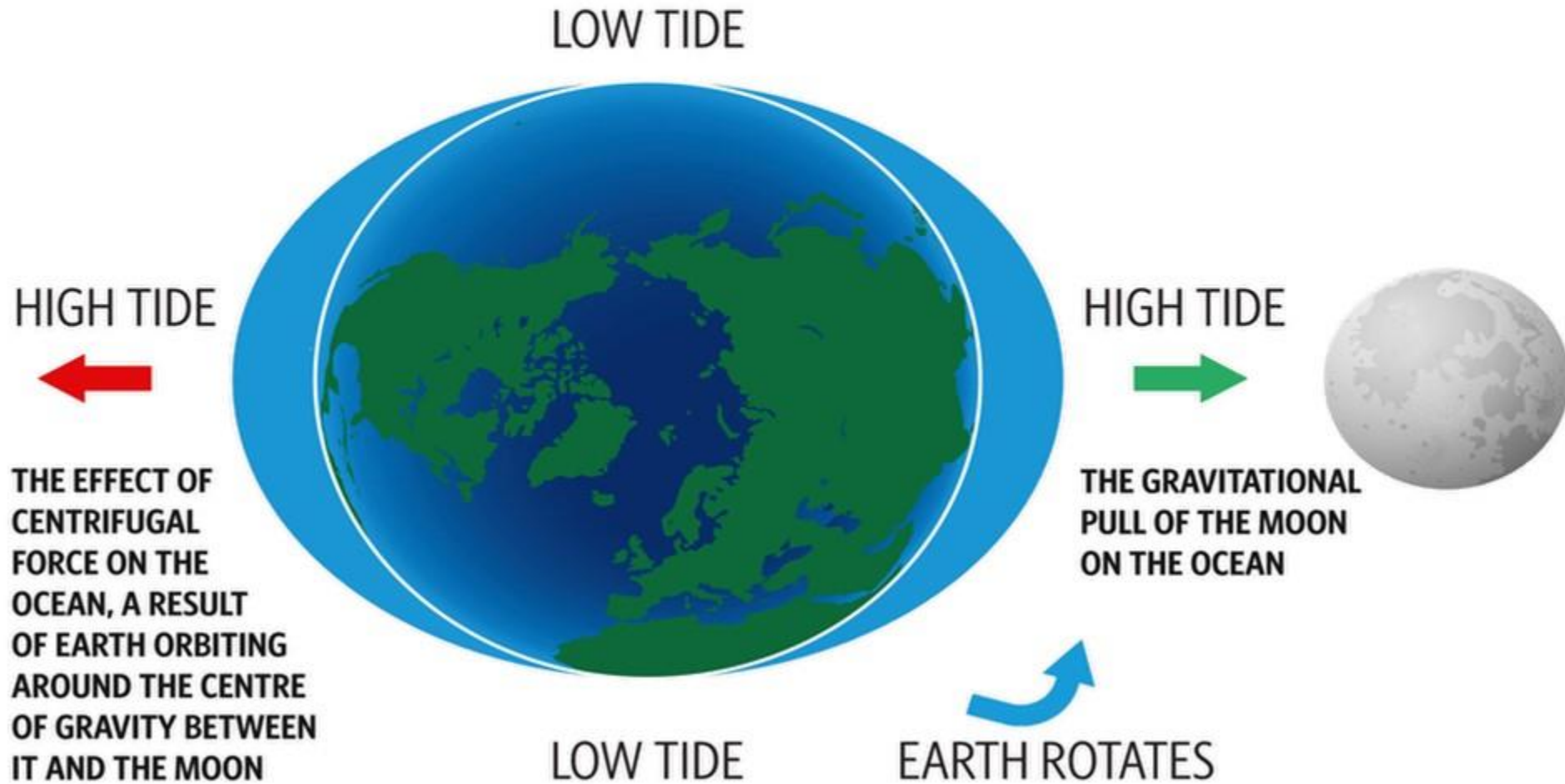
3 On the opposite side though the gravitational attraction of the moon is less as it is farther away, the **centrifugal force causes tidal bulge on the other side**.

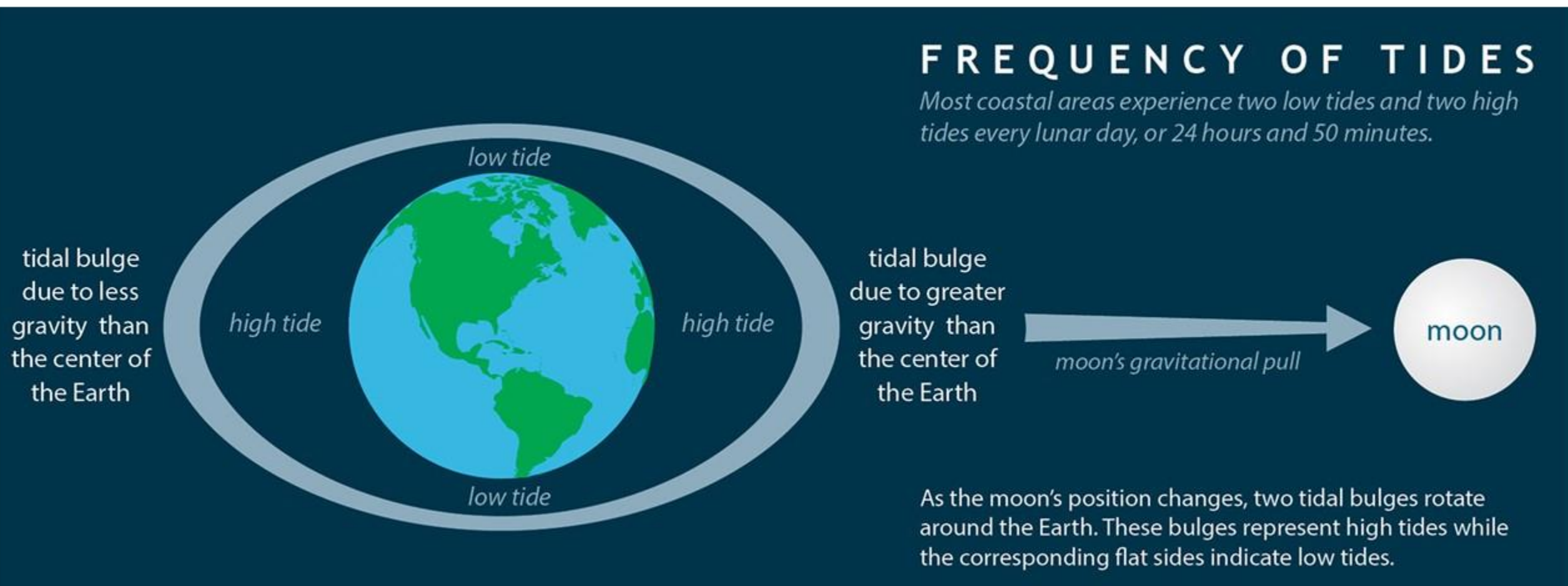
Why is it so?

4 On the opposite side of the earth, the attractive force is less, as it is farther away from the moon, **the centrifugal force is dominant**. Hence, there is a net force away from the moon.

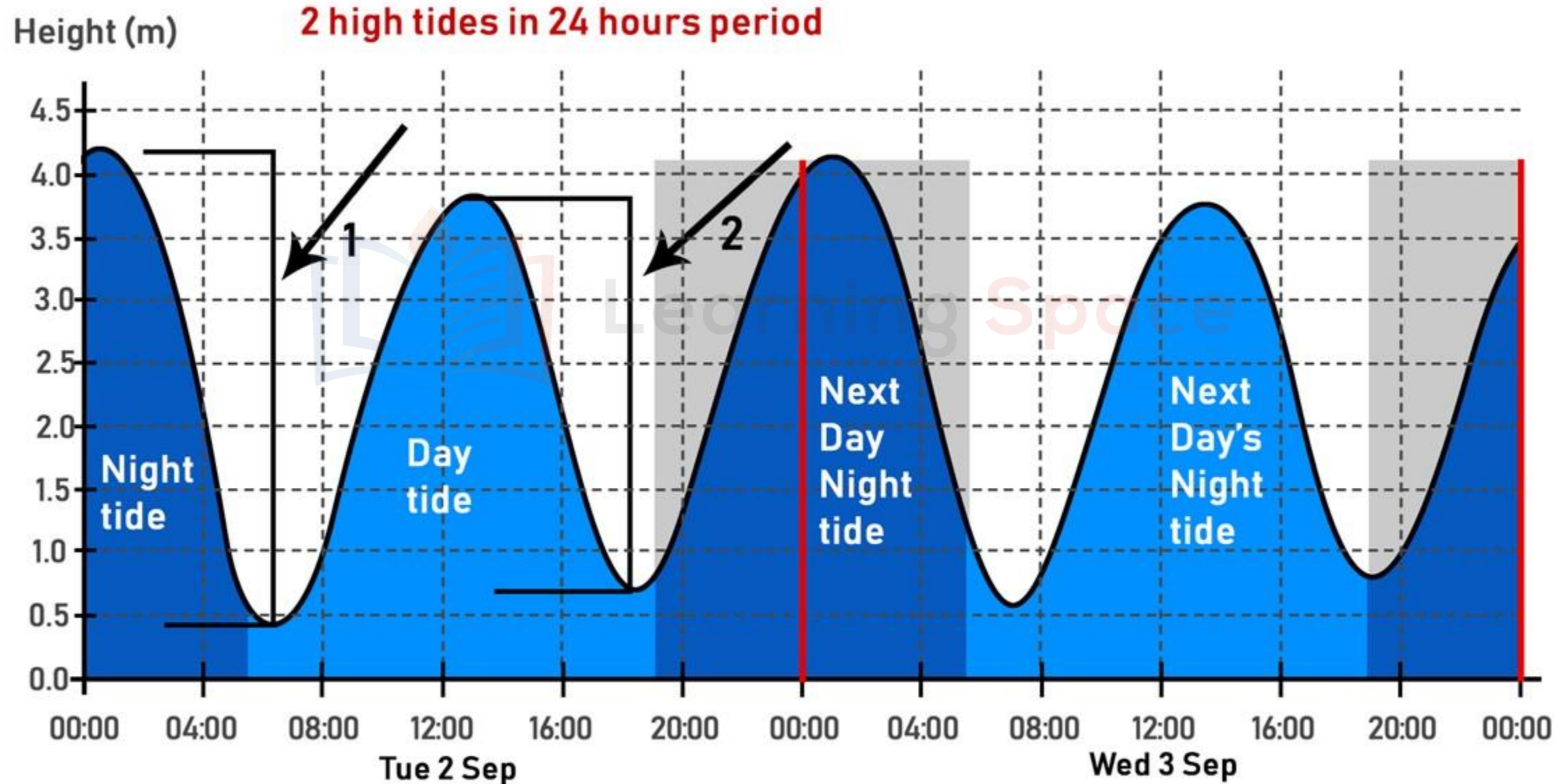
The 'tide-generating' force is the absolute difference between these two forces; i.e. the gravitational attraction of the moon and the centrifugal force.

FORCES THAT INFLUENCE THE TIDES





MOVEMENTS OF OCEAN WATER



MOVEMENTS OF OCEAN WATER

TIDES

DOES THE SHAPE OF COASTS INFLUENCE TIDES?

- 1 The tidal bulges on wide continental shelves, have greater height.



- 2 When tidal bulges hit the mid-oceanic islands they become low.
- 3 The shape of bays and estuaries along a coastline can also magnify the intensity of tides.
- 4 Funnel-shaped bays greatly change tidal magnitudes.
- 5 When the tide is channelled between islands or into bays and estuaries they are called tidal currents.



TIDES OF BAY OF FUNDY, CANADA

- The highest tides in the world occur in the Bay of Fundy in Nova Scotia, Canada.
- The tidal bulge is 15 - 16 m.
- Because there are two high tides and two low tides every day (roughly a 24 hour period); then a tide must come in within about a six hour period.
- As a rough estimate, the tide rises about 240 cm an hour.
- If you have walked down a beach with a steep cliff alongside (which is common there), make sure you watch the tides.

WE EXPRESS OUR SINCERE THANKS FOR VIEWING THIS VIDEO

Presented by



Learning Space

For Suggestions:

suggestions@learningspace.in

To Contact us:

info@learningspace.in

OUR TEAM

G. V. Rao
K. Srikanth
D. Sunil Kumar
L. V. Krishna
Y. Deepthi

Amrita Naidu
M. Binukrishna
G. Ravi Babu
D. Joji
K. Victor Babu

Visit us at:

www.learningspacedigital.com

0 8 6 6 -2 4 4 4 4 7 2
0 9 8 4 9 9 4 2 2 9 9